Protocol Entry 006: Generator 3 Primary Memory Confirmation Closure

BH Stability Vector Space Protocol Record

2026-07-21

Question

Does a strictly causal self-history feature improve next-step λ_{bh} prediction beyond the frozen current-state and cadence baseline under a selection-aware, within-track null?

Frozen design

The baseline was $[\lambda_t, \Delta u_{t\to t+1}]$. The candidate grid was $\tau/T \in \{0.02, 0.05, 0.10, 0.20\}$. Memory was recomputed after within-track-and-split permutation and kernel selection was repeated inside every null replicate. The confirmation used 2,000 null replicates and the pre-frozen 300,000-row, 400-track sample.

Result

Quantity	Value
Observed fractional RMSE improvement	0.042627
Null mean fractional improvement	0.026412
Null-adjusted order excess	0.016215
Null 95th percentile	0.027086
Exceedances	0/2,000
Plus-one one-sided p	0.0004998

The frozen confirmation passed. The null-adjusted excess is approximately 1.62%, above the declared 1% practical floor. Most of the raw improvement remains null-preserved denoising and is not labeled memory.

Adjudication

The authorized statement is: *a small order-dependent self-history predictive gain exists in the frozen TNG λ-track primary design beyond the current state and cadence baseline.* This closes the pre-registered existence test. No further permutations of the same design are authorized.

The result does not confirm structure beyond AR(3), identify a physical timescale, establish independent H/S axes, establish host regulation, or admit HRSM M. Post-hoc AR(3) diagnostics remain separate and show scale-fragile, short-scale order dependence with unresolved fixed-row depth.

Provenance

Raw confirmation SHA-256: `f3098426b7181802aa7ce9c08ab6411a2ed4d0a8dfb3e37148f30f246cc00c51`
Freeze manifest SHA-256: `dfbba783a75e02da206ed2c1dc42bf39a1612738fa5dc3c0720e228795901776`